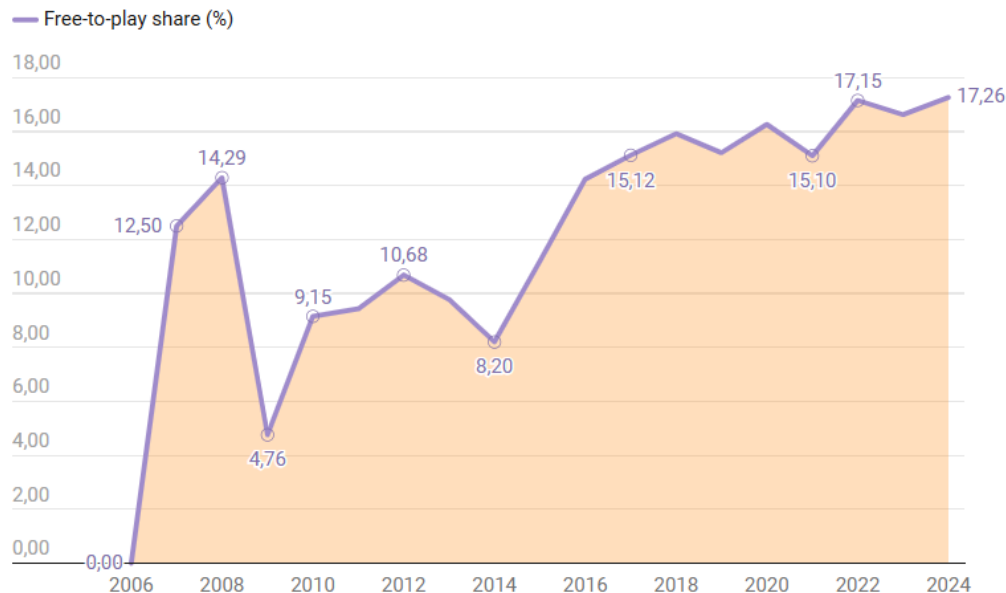


Data Visualization Pitch

Student Individual Assignment

The rise of free-to-play games on Steam (2006–2024)



Games That last: A New Era of Video Games on Steam

Business models, game design, and
player engagement in the video
game industry

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<https://github.com/mroscioli/VideoGames/tree/main>

Research questions

How do business models shape success and player engagement on Steam?

- ❑ Do free-to-play games reach larger audiences than paid games on Steam?
- ❑ How do free-to-play and paid games differ in terms of player engagement and activity?
- ❑ Are free-to-play games associated with different patterns of player satisfaction compared to paid games?
- ❑ How has the adoption of free-to-play models evolved over time on Steam?

What makes a game last on Steam over time?

- ❑ How does game design (single-player, multiplayer, or hybrid) influence long-term player engagement?
- ❑ Do multiplayer and hybrid games show different patterns of longevity compared to single-player games?
- ❑ How does recent player activity relate to historical success and popularity?

About data

During the data exploration phase, several considerations emerged regarding data availability, completeness, and the interpretation of engagement metrics. In particular, understanding player activity over time required careful evaluation of how engagement-related variables were recorded and represented in the dataset.

Datasets

Steam Games Dataset 2025 (Kaggle)

The dataset provides detailed metadata for nearly 90,000 games released on Steam, including pricing information, estimated ownership ranges, player reviews, playtime statistics, release years, and categorical descriptors such as genres and gameplay modes. Steam was chosen as the analytical context because it represents one of the largest and most data-rich digital PC game platforms, making it well suited for studying player engagement and longevity.

Data quality and limitations

Several limitations were identified:

- ❑ Ownership figures are provided as estimated ranges rather than exact values, requiring the use of midpoint approximations.
- ❑ Playtime variables contain a large number of zero values, which reflect missing or unrecorded data rather than true zero engagement.
- ❑ Engagement metrics (reviews, playtime, activity) represent only the Steam ecosystem.
- ❑ Some categorical fields (e.g., genres and gameplay features) exhibit inconsistent labeling, requiring careful interpretation.

Anticipated data cleaning operations

- ❑ Selection of a subset of relevant variables aligned with the research questions.
- ❑ Conversion and standardization of numerical fields to ensure consistent data types.
- ❑ Derivation of proxy measures, such as ownership tiers based on estimated owner ranges.
- ❑ Classification of games into gameplay modes based on available categorical information.
- ❑ Restricting certain analyses to subsets of games with sufficient data availability (e.g., playtime metrics).

Licensing and data usage

The dataset is released under the MIT License, which allows reuse, modification, and redistribution for academic and non-commercial purposes, provided proper attribution is maintained. This makes it suitable for the intended educational use.

Methodology

Data Processing & Cleaning

All data processing was conducted using Python, primarily relying on the pandas and NumPy libraries. Initial cleaning steps included:

- ❑ Selection of a subset of variables relevant to the research questions.
- ❑ Conversion of numerical fields to consistent data types.
- ❑ Handling of missing and ambiguous values (e.g., zero values in playtime variables interpreted as unrecorded data rather than zero engagement).

Data Collection

The analysis is based on a publicly available dataset downloaded manually from Kaggle.

The Steam Games Dataset aggregates metadata and engagement-related information collected from the Steam platform, including pricing, estimated ownership ranges, player reviews, playtime statistics, and categorical descriptors. No web scraping or API queries were performed, ensuring consistency and transparency in data sourcing.

Reproducibility

The analysis is fully reproducible using the original dataset. All transformations and analytical steps are deterministic and based on publicly available data, allowing the study to be replicated or extended by other researchers using the same data source.

Data Processing & Cleaning

- ❑ Estimated ownership ranges were converted into midpoint values and grouped into ownership tiers.
- ❑ Games were classified into single-player, multiplayer, and hybrid categories based on genre and feature metadata.
- ❑ Derived indicators were created to distinguish free-to-play and paid games based on pricing information.
- ❑ Temporal variables were extracted from release dates to enable time-based comparisons.

Insights from the Data

1) Game success on Steam is highly concentrated

The majority of games fall into the lowest ownership tiers, while only a very small fraction reach millions of players.

This long-tail distribution highlights the competitive nature of the platform and the difficulty of sustaining large audiences.

2) Free-to-play games tend to reach broader audiences than paid games

Free-to-play titles are overrepresented among the highest ownership tiers and are more likely to accumulate large volumes of player reviews, indicating higher reach and visibility.

3) Paid games generally show higher and more consistent player satisfaction

While free-to-play games attract more players, paid games are more frequently associated with higher percentages of positive reviews, suggesting different engagement and expectation dynamics.

4) The adoption of free-to-play models has increased over time

Free-to-play games represent a growing share of new releases on Steam, supporting the idea of a structural shift in business models across recent years.

5) Game design plays a key role in long-term player engagement.

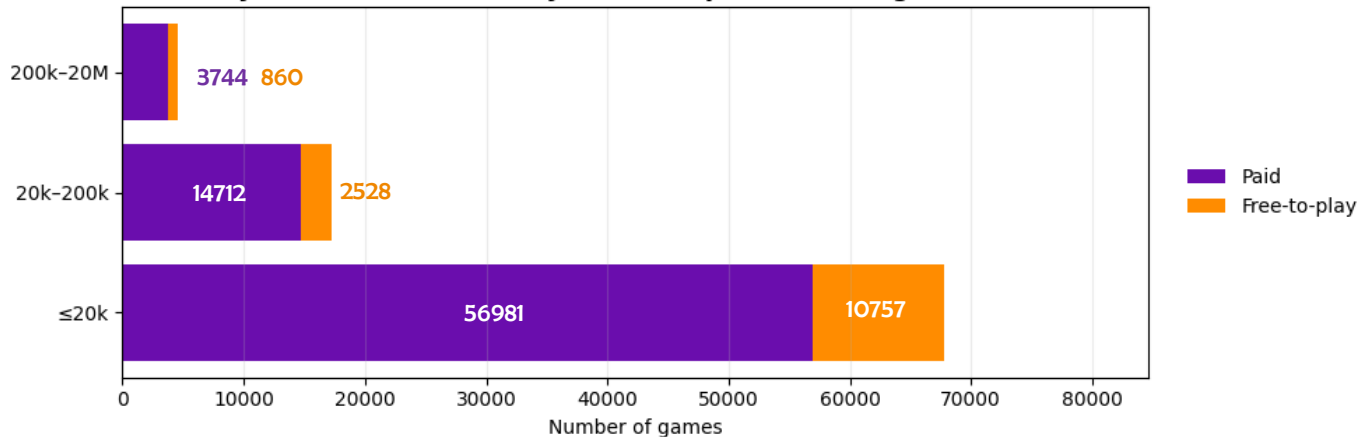
Hybrid games combining single-player and multiplayer features are more likely to sustain engagement over time, while multiplayer-only games exhibit higher intensity of recent playtime when active. Single-player games, although more numerous, are less likely to maintain ongoing engagement.

Analysis Operations Performed

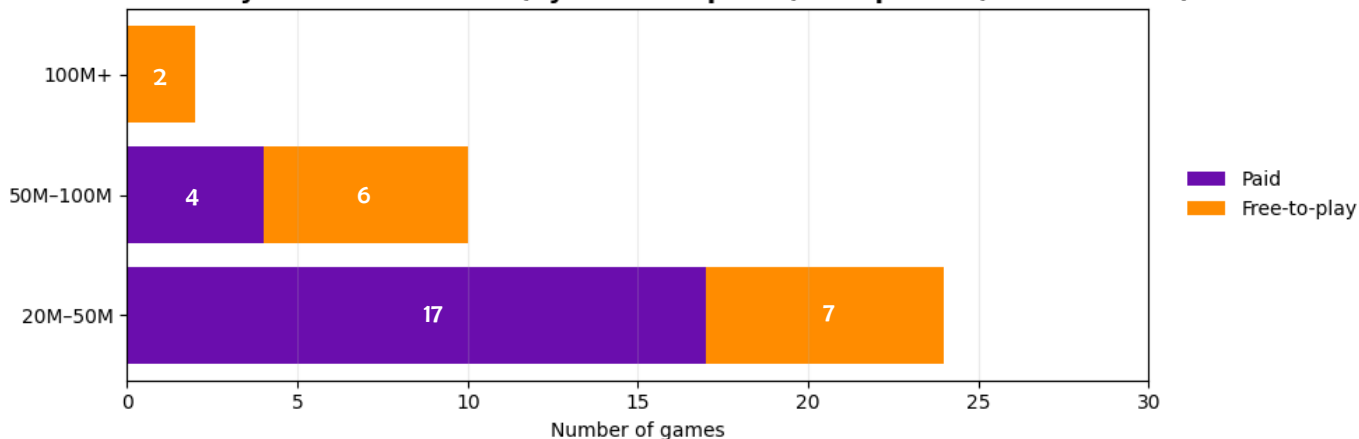
- Descriptive statistics were used to examine distributions, medians, percentiles, and extreme values for ownership, reviews, ratings, and playtime.
- Comparative analysis was conducted to contrast free-to-play versus paid games, as well as single-player, multiplayer, and hybrid game designs.
- Filtering and stratification were applied to focus on subsets of games with sufficient data availability (e.g., ownership thresholds for playtime analysis).
- Temporal analysis was used to observe changes in pricing models and engagement patterns across release periods.
- Derived indicators (ownership tiers, gameplay mode classifications, availability of playtime data) were created to support consistent comparisons.

Free-to-play vs Paid: Audience Reach and Market Concentration

Free-to-Play vs Paid on Steam:(by Ownership Tier) — Long tail ($\leq 20M$ owners)



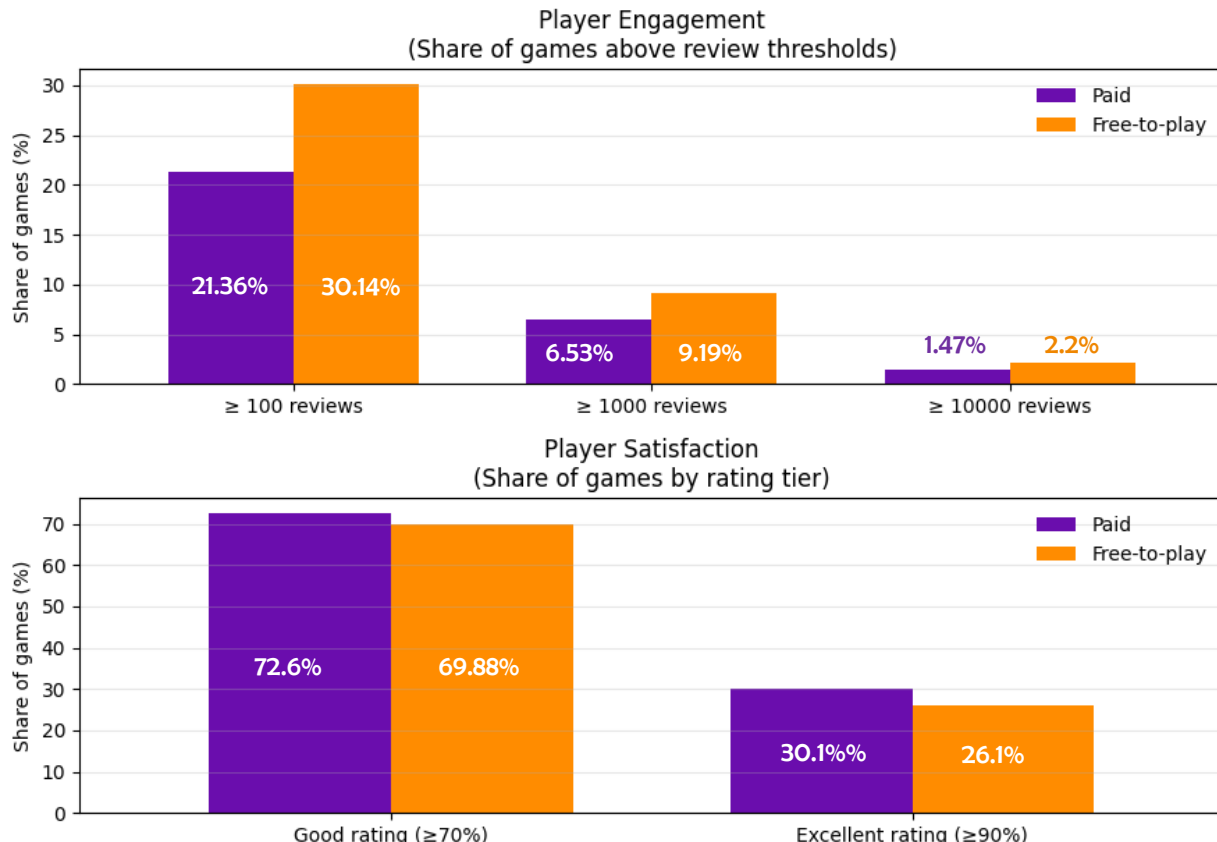
Free-to-Play vs Paid on Steam:(by Ownership Tier) — Top tiers ($\geq 20M$ owners)



- ❑ The majority of Steam games belong to the long tail of low ownership and are mostly paid titles.
- ❑ Free-to-play games are relatively rare overall, but become increasingly common in the highest ownership tiers.
- ❑ At the top of the market, free-to-play titles are disproportionately represented compared to paid games.

This indicates that free-to-play models are more likely to achieve very large player bases, despite being fewer in number.

Engagement and Satisfaction by Business Model



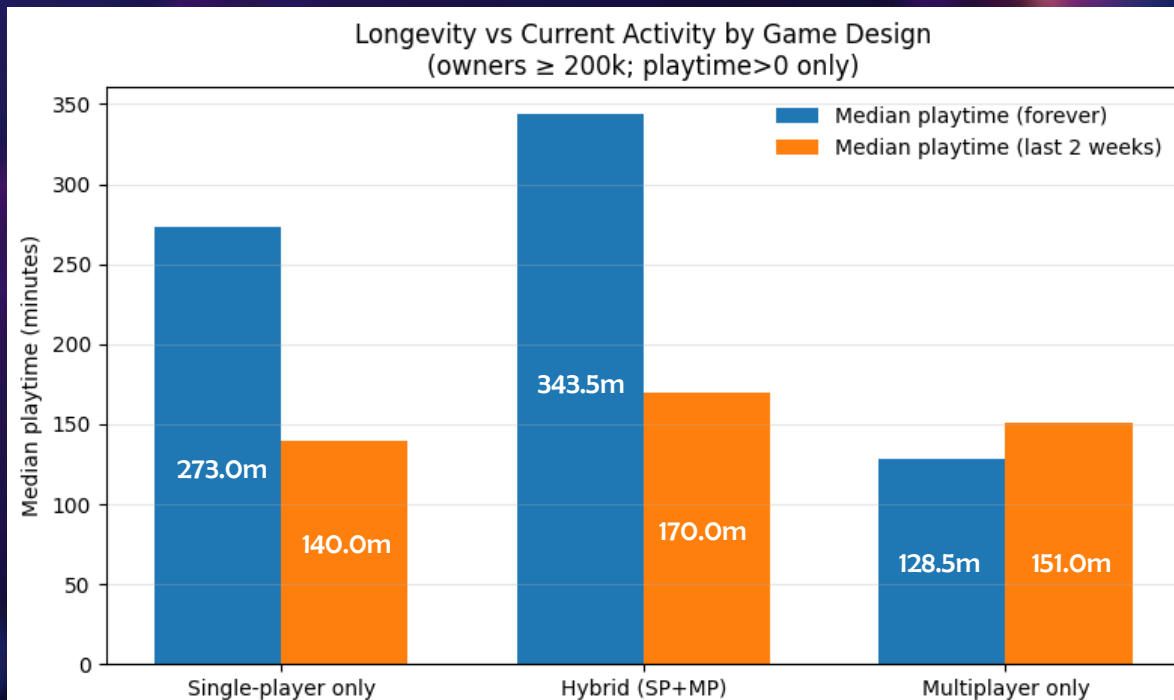
- Free-to-play games show consistently higher levels of player engagement, being more likely to exceed high review thresholds (≥100, ≥1k, ≥10k reviews).
- Paid games are less represented at high engagement levels, reflecting a more limited but stable audience reach.
- When focusing on player satisfaction, both business models perform similarly: comparable shares of free-to-play and paid games achieve good (≥70%) and excellent (≥90%) ratings.

Overall, free-to-play models amplify engagement and visibility, while perceived player satisfaction remains largely independent of the business model.

Longevity vs Current Activity by Game Design

- ❑ Game design plays a crucial role in shaping how long games remain engaging on Steam, both historically and in the present.
- ❑ Hybrid games (single-player + multiplayer) emerge as the most resilient design: they accumulate the highest median playtime over their lifetime and maintain relatively strong recent activity.
- ❑ Single-player games show solid long-term engagement, indicating that players invest significant time after release, but their recent activity is noticeably lower, suggesting a tendency toward finite or completed experiences.
- ❑ Multiplayer-only games, while accumulating less total historical playtime on average, show a smaller gap between lifetime and recent activity, indicating a design more oriented toward continuous engagement.

Taken together, these patterns suggest that combining structured single-player content with ongoing multiplayer interactions is the most effective strategy for sustaining player engagement over time on Steam.



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by Manuel Roscioli

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